Xiaoyu Zhang

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Research Summary

I build **multimodal sensing agents** that *perceive*, *reason*, and *act* on the physical world by jointly leveraging sensing modalities organized across four complementary physical domains: **radio-frequency** (mmWave FMCW radar, synthetic-aperture and cascaded-array imaging), **acoustic / ultrasonic** (photoacoustic imaging, ultrasound, audio and speech), **optical / radiographic** (visible-light vision, infrared, mammography), and **bio-electrical** (wearable ear-EEG). My research closes the loop from **novel sensing front-ends** through **physics-informed inverse imaging** and **cross-modal representation learning** to **deployed clinical systems**, with validated impact on **smart health** (wound care, breast cancer subtyping, thyroid diagnosis, Alzheimer's screening, hearing disorders) and **social good** (sustainable building, accessible HCI, point-of-care diagnostics). My work has produced **15 peer-reviewed papers** (7 first-authored, at venues including ACM IMWUT, ACM UIST, IEEE IoTJ, IEEE/RSJ IROS, and npj Imaging), a **Best Paper Candidate** at ACM SenSys HumanSys'25, **4 patents** (one under commercialization by a medical-device startup, whose technology I pitched to the Panasci TEC 2025 semifinals), and a **clinical station deployed under an IRB-approved validation study**, reflecting a deliberate *research–clinic–industry* translation pipeline.

Research Interests

My research lies at the intersection of **Wireless Sensing**, **Multimodal Medical AI**, and **Embodied Intelligence**. One technical question recurs across all three: *when the sensor is cheap and the data are scarce, where should the burden of perception sit — on physics, on learning, or on the hardware aperture?*

1) mmWave Sensing and Imaging. Turning low-cost commodity radar into a high-resolution instrument for sensing the human body, by shifting the burden of resolution onto embedded physics, learned cross-modal supervision, or a synthesized aperture.

- *Clinical imaging* — wound assessment through dressings; tissue moisture and perfusion mapping
- *Human–machine interaction* — hand and gesture tracking for touchless control in sterile or low-light settings
- *Identity and affect* — mmWave–voice multi-factor identification; micro-expression sensing
- *Privacy-preserving monitoring* — presence and activity sensing that carries no identifying imagery

2) Multimodal Medical AI. Fusing and transferring across imaging and physiological signals where labeled clinical data are scarce, and testing whether a pretrained foundation model *genuinely* transfers to a new clinical population.

- *Oncology* — non-invasive breast-cancer molecular subtyping from photoacoustic, ultrasound and mammographic imaging
- *Endocrine* — unified thyroid nodule diagnosis and lesion segmentation from clinical ultrasound
- *Wound and vascular* — chronic foot-ulcer prognosis and perfusion prediction from photoacoustic imaging
- *Neurological and auditory* — Alzheimer's subtype screening from speech; sound-sensitivity screening from wearable ear-EEG

3) Robotic Perception and Autonomous Inspection. Equipping autonomous platforms to assess material and structural condition non-destructively, and to close a control loop on that assessment.

- *Building energy* — autonomous robotic inspection of thermal insulation beneath building surfaces

- *Sustainable manufacturing* — closed-loop visual process control for construction-material production

Education

University at Buffalo, the State University of New York (SUNY) *Sep. 2021 - Jul. 2026*
Ph.D., Computer Science and Engineering Co-advised by Prof. Wenyao Xu & Yaxiong Xie
Dissertation (**defended Jul. 2026**): *Toward High-Resolution and High-Capacity mmWave Sensing Technology for Medical Applications*
Committee: Wenyao Xu, Yaxiong Xie, Jun Xia

University of Science and Technology of China *Sep. 2017 - Jun. 2020*
M.S., Electronics and Communication Engineering Advised by Prof. Bin Liu

Hefei University of Technology *Sep. 2013 - Sep. 2017*
B.Eng., Electronic Information Engineering

Publications

Summary. 15 peer-reviewed papers, **7 first-authored** (name in **bold**), spanning flagship venues in ubiquitous computing (ACM IMWUT, UIST), robotics (IEEE/RSJ IROS), medical imaging (Photoacoustics, npj Imaging), and biomedical informatics (IEEE JBHI, IoTJ). Thrust tags: [Health] [Embodied] [Social Good].

[IROS'26] [Social Good] **Xiaoyu Zhang**, Ye Zhan, Chenhan Xu, Zhengxiong Li, Chuqin Huang, Yanda Cheng, Licheng Liang, Wei Bo, Jun Xia, Chi Zhou, Hongyue Sun, Wenyao Xu, "*Autonomous Robotic Inspection of Building Thermal Insulation via Subsurface mmWave Sensing*", IEEE/RSJ International Conference on Intelligent Robots and Systems, 2026. *(Patent filed.)*

[IMWUT'26] [Health] Shuwei Hou, Wei Bo, L. Cao, Varun Shijo, Chuhui Liu, **Xiaoyu Zhang**, L. Guo, Wenyao Xu, "*Speech Annotation and Transcription Enhancer (SATE): An Automated System for Child Language Sample Analysis*", Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 2026.

[ICIBM'26] [Health] **Xiaoyu Zhang**, Wei Bo, Anarghya Das, Longxiang Pan, Wenyao Xu, "*Matching Speech Models to ADRD Subtypes: A Mechanistic Benchmark*", International Conference on Intelligent Biology and Medicine, 2026.

[npj Imaging'26] [Health] Yanda Cheng, Chuqin Huang, Shuliang Yu, Saptarshi Chakraborty, Yunqi Xi, Robert W. Bing, Huijuan Zhang, **Xiaoyu Zhang**, Isabel Komornicki, Linda M. Harris, Wenyao Xu, Jun Xia, "*Predictive modeling of chronic foot ulcer outcomes using longitudinal photoacoustic imaging*", npj Imaging, 2026.

[HealthCom'26] [Health] N. Wei, A. Xu, **Xiaoyu Zhang**, "*Pool or Specialize? Anatomic-Site Domain Shift in Melanoma Classification*", IEEE International Conference on E-health Networking, Application & Services, 2026.

[IoTJ'25] [Health] **Xiaoyu Zhang**, Zhengxiong Li, Yanda Cheng, Chenhan Xu, Chuqin Huang, Emma Zhang, Ye Zhan, Wei Bo, Jun Xia, Wenyao Xu, "*mmSkin: An Over-gauze Wound Assessment System using Radio Frequency Technologies*", IEEE Internet of Things Journal, 2025. *(Translated into US patent; under commercialization by a medical-device startup.)*

[IoTJ'25] [Embodied] **Xiaoyu Zhang**, Zhengxiong Li, Chenhan Xu, Luchuan Song, Huining Li, Hongfei Xue, Yingxiao Wu, Wenyao Xu, "*mmHand: Towards Pixel-Level-Accuracy Hand Localization Using a Single Commodity mmWave Device*", IEEE Internet of Things Journal, 2025.

[HumanSys'25] [Health] **Xiaoyu Zhang**, Zhengxiong Li, Yanda Cheng, Chenhan Xu, Chuqin Huang, Emma Zhang, Ye Zhan, Wei Bo, Jun Xia, Wenyao Xu, "*Through-dressing Wound Monitoring Based on* *the mmWave Sensor*", The 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys workshop, co-located with ACM SenSys), 2025. (**Best Paper Candidate**)

- **[Sensors'25]** [Health] **Xiaoyu Zhang**, Chuhui Liu, Yanda Cheng, Zhengxiong Li, Chenhan Xu, Chuqin Huang, Ye Zhan, Wei Bo, Jun Xia, Wenyao Xu, "*A Comprehensive Survey of Research Trends in mmWave Technologies for Medical Applications*", Sensors, 2025.
- **[Photoacoustics'25]** [Health] Chuqin Huang, Yanda Cheng, **Xiaoyu Zhang**, Ye Zhan, Wenhan Zheng, Isabel Komornicki, Linda M. Harris, Wenyao Xu, Jun Xia, "*Radiomics-Driven Perfusion Prediction in Clinical Photoacoustic Foot Imaging*", Photoacoustics, 2025.
- **[Photoacoustics'25]** [Health] Chuqin Huang, Emily Zheng, Wenhan Zheng, Huijuan Zhang, Yanda Cheng, **Xiaoyu Zhang**, Varun Shijo, Robert W. Bing, Isabel Komornicki, Linda M. Harris, Ermelinda Bonaccio, Kazuaki Takabe, Emma Zhang, Wenyao Xu, Jun Xia, "*Enhanced clinical photoacoustic vascular imaging through a skin localization network and adaptive weighting*", Photoacoustics, 2025.
- **[JBHI'24]** [Health] Wei Bo, Suzanne S. Sullivan, **Xiaoyu Zhang**, Mingchen Gao, Wenyao Xu, "*A Telemedicine Analytic Framework for Fully and Semi-automatic Alzheimer's Disease Screening using Clock Drawing Test*", IEEE Journal of Biomedical and Health Informatics, 2024.
- **[UIST'23]** [Embodied] Tiantian Liu, Feng Lin, Chao Wang, Chenhan Xu, **Xiaoyu Zhang**, Zhengxiong Li, Wenyao Xu, Ming-Chun Huang, Kui Ren, "*WavoID: Robust and Secure Multi-modal User Identification via mmWave-voice Mechanism*", ACM Symposium on User Interface Software and Technology, San Francisco, USA, 2023.
- **[BSN'19]** [Embodied] **Xiaoyu Zhang**, Bin Liu, "*A Channel Hopping Strategy Based on the Human Trajectory Similarity for WBANs*", IEEE-EMBS International Conference on Body Sensor Networks, Chicago, USA, 2019.
- **[Bodynets'17]** [Embodied] Chengjie Guan, Bin Liu, Zhiqiang Liu, Y. Zhang, **Xiaoyu Zhang**, "*JMMM: A Mobility Model for WBANs Based on Human Joint Movements*", 19th EAI International Conference on Body Area Networks, Dalian, China, 2017.

Honors and Awards

- **Best Paper Candidate Award**, HumanSys'25 — The 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (co-located with ACM SenSys), 2025
- **Semifinalist**, Panasci Technology Entrepreneurship Competition (Panasci TEC), University at Buffalo, 2025
- **Graduate Teaching Award**, Department of CSE, University at Buffalo, 2024
- **Chair's Fellowship**, Department of CSE, University at Buffalo, 2021
- Merit scholarships (graduate and undergraduate), USTC and HFUT, 2015–2019

Industry & Clinical Experience

Clinical Research Collaborations *2022 - Present*

Multi-year IRB-approved clinical collaborations at the University at Buffalo and partner sites, spanning device deployment, on-site data collection, and multi-modal cohort curation:

- **Wound care** (UBMD Vascular Surgery) — deployed the WaveSight-RF clinical station on site; **5 patients** scanned to date under an IRB-approved protocol, with repeat longitudinal scanning up to **3 months** per patient
- **Breast imaging** — clinical cohort of **284 imaged patients**; curated the **53-patient** tri-modal photoacoustic / ultrasound / mammography subset used for molecular subtyping
- **Wearable ear-EEG — 27 subjects** recorded across four passive auditory tasks (∼1,800 trials), **23** retained after per-channel quality control, paired with standardized clinical score scales
- **Thyroid ultrasound** — ongoing clinical scanning collaboration for nodule diagnosis and lesion segmentation

- **Photoacoustic Imaging Lab** (Dept. of Biomedical Engineering) — joint imaging-system development and radiomics analysis

China Merchants Bank Software Center *Jul. 2020 - Sep. 2020*
Designed and implemented a full-stack data-verification system (Bootstrap + Django) with dynamic front-back interaction for multi-source financial data.

Competition

- **Panasci Technology Entrepreneurship Competition (Panasci TEC) 2025** *Mar. 2025*
  Advanced Through-Gauze Wound Assessment Technology.
  Xiaoyu Zhang (technical lead and presenter), Chuhui Liu, Jun Xia, Linda M. Harris, Wenyao Xu.
  Advanced to the **semifinals**; pitched the patented mmWave wound-assessment platform to clinical and industry judges.

Presentations & Demos

- **Ph.D. Dissertation Defense**: *Toward High-Resolution and High-Capacity mmWave Sensing Technology for Medical Applications* Jul. 2026
  University at Buffalo, CSE
- **Live Demonstration**: SUNY Research Expo Mar. 2026
  WaveSight-RF clinical wound-imaging station — translational outreach
- **Community Engagement Studio**: UB CTSI Feb. 2026
  Wound-measurement device design review with a 12-member community expert panel
- **Invited Talk**: The 3rd International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys, co-located with ACM SenSys) May 2025
  Through-dressing Wound Monitoring Based on the mmWave Sensor
- **Pitch Talk**: Panasci Technology Entrepreneurship Competition Mar. 2025
  Advanced Through-Gauze Wound Assessment Technology
- **Invited Talk**: IEEE-EMBS International Conference on Body Sensor Networks May 2019
  A Channel Hopping Strategy Based on the Human Trajectory Similarity for WBANs
- **Invited Talk**: 19th EAI International Conference on Body Area Networks Sep. 2017
  JMMM: A Mobility Model for WBANs Based on Human Joint Movements

Teaching Experiences

Teaching Assistant across **6 semesters** at the University at Buffalo, including one invited guest lecture, plus one semester at USTC.

1. Algorithm Analysis and Design, UB *Fall 2021, Spring 2022, Fall 2022, Fall 2023, Fall 2024*
2. Special Topics, UB — incl. **invited guest lecture**: *Wireless Signal Processing: Making Sense of the Invisible* *Fall 2023*
3. Mathematical Logic and Graph Theory, USTC *Fall 2018*

Mentoring Experiences

I have mentored **10+ students** on wireless sensing, embedded systems, and AI-based medical screening projects.

- Prakriti Setlur — AI-based clinical image screening pipelines
- Kh Shafin Farhan — foundation models for medical imaging
- Abhi Ramtel (CSE@UB) — UI design of an over-gauze wound assessment system
- Cole Desimone (AE@UB) — 3D model design of the mmWave sensor scanning system
- George Gillman (EE@UB) — mmWave technologies for medical applications: a review
- Weida Jiang (CSE@UB) — embedded mmWave-based hand detection on Raspberry Pi
- Wenxuan Huang & Yiwen Tan (CSE@UB) — tooling for automatic image labeling

- High-school summer research interns — AI-based medical screening pipelines, mentored through UB outreach programs

Community Services & Outreach Activities

Conference Organization:

- **Organizing Committee Member (Local Arrangements)**, International Conference on Intelligent Biology and Medicine (ICIBM) [2026]

Journal Reviewer:

- IEEE Transactions on Mobile Computing (TMC) [2025, 2026]
- ACM Transactions on Computing for Healthcare (HEALTH) [2025]
- IEEE Open Journal of Engineering in Medicine and Biology (OJEMB) [2025]
- Scientific Reports [2025]
- Smart Health (SH) [2024, 2025]

Conference Reviewer:

- ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE) [2025]
- IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI) [2024, 2026]
- Physics Education Research Conference (PERC), American Association of Physics Teachers (AAPT) [2026]
- IEEE-EMBS International Conference on Body Sensor Networks (BSN) [2023, 2024]

Outreach:

- Live technology demonstration, **SUNY Research Expo** — translational outreach for the WaveSight-RF clinical wound-imaging station [Mar. 2026]
- **Community Engagement Studio**, UB Clinical and Translational Science Institute (CTSI) — convened a **12-member community expert panel** (healthcare providers, clinical trainees, community advocates) for a design review of the wound-measurement device [Feb. 2026]